

Master's student in Computer Science with experience building scalable backend systems, AI-powered applications, and full-stack platforms. Strong foundation in data structures, distributed systems, cloud-based ML pipelines, and software architecture. Experienced in Python, Java, JavaScript, modern web frameworks, and AI infrastructure with hands-on work in automation, agentic AI systems, computer vision, and production-grade engineering workflows.

TECHNICAL SKILLS

Languages	Python, Java, C/C++, JavaScript, SQL
Frameworks & Libraries	PyTorch, TensorFlow, scikit-learn, OpenCV, LangChain, LangGraph, CrewAI, Agno, Flask, Django, React
Tools & Platforms	Git, GitHub, Jupyter Notebook, Google Colab, VS Code, MCP Servers, LaTeX, Figma

EXPERIENCE

Teaching Assistant

Jan 2026 – Present

Data-Driven WV – Cloud Analytics / NOAA Data

- Prepared and structured NOAA-based datasets and project pipelines (GOES-16, weather data) ahead of the course, reducing student setup time by 30% and enabling smooth onboarding.
- Designed scalable data workflows and spatiotemporal ML tasks, improving project performance and evaluation consistency by 25%.
- Developed documentation, starter code, and debugging guides, boosting overall project quality and submission success rate by 35%.

Graduate Research Assistant

Aug 2025 – Present

AI/ML/Deep Learning

- Built end-to-end object detection and XAI pipelines using YOLOv5, trained and evaluated on a multi-class, hand-labeled satellite image dataset.
- Designed a quantitative Q-metric for XAI evaluation that reduced manual heatmap inspection by 70%, enabling faster identification of faithful and trustworthy explanation methods.
- Developed a full-stack web application for the YOLOv5-XAI pipeline that reduced experiment execution and iteration time by 50%, eliminating repeated manual code runs.
- Developed a video-based surgical analysis pipeline using SAM3 segmentation on 20 laparoscopic sleeve gastrectomy videos, enabling automated extraction of anatomical regions with improved consistency.
- Designed a scaling-based normalization technique to address frame inconsistencies, achieving 30% more reliable cross-frame pixel comparisons for surgical transformation analysis.
- Computed quantitative cut-ratio metrics across all videos (including surgeon-wise analysis), reducing manual evaluation effort by 40% and enabling objective surgical outcome assessment.

Smart Interviews

Jan 2024 – May 2024

Instructor & Mentor

- Taught core DSA topics including arrays, strings, linked lists, stacks, and hash tables to a cohort of 80+ students.
- Conducted doubt-solving sessions and coding practice to prepare students for technical assessments.
- Played a key role in enhancing students' problem-solving abilities and interview readiness through continuous support and feedback.

PROJECTS

Multi-Agent Debate RAG System

- Designed a multi-agent debate-based RAG system using LangGraph and Ollama, reducing weakly grounded or hallucinated responses by 45% compared to a single-agent RAG baseline.
- Implemented cross-examination and judge synthesis between independent agents, improving answer completeness and evidence coverage by 30%.
- Built an automated citation-verification pipeline enforcing 100% citation coverage for factual claims and eliminating uncited statements in final responses.

EDUCATION

West Virginia University, Morgantown, WV

M.S. in Computer Science

Jan 2025 – Dec 2026 (Expected)

Relevant Coursework: Agentic AI, Pattern Recognition, Deep Learning, Big Data

GPA: 3.87

IBM RAG and Agentic AI

Professional Certificate

Sep 2025